SARDAR VALLABHBHAI NATIONAL INSTITUTE OF TECHNOLOGY, SURAT
DEPARTMENT OF ARTIFICIAL INTELLIGENCE
B. Tech-4th Year (7th Sem)
Introduction to LLM(AI463)
Mid Semester Exam

Date: 25/09/2026 Time: 09:00 AM To 10:30 AM Marks: 6*5=30

1. What is Word2Vec and how is it different from TF-IDF representation? Show Word2Vec's architecture (any one). What is the difference in formulation between Skip-gram without negative sampling and Skip-gram with negative sampling? [CO2] [2+2+2=6]

2. You are designing neural language model. The vocabulary is restricted to V={ cat, sat, mat, </s>}, where </s> represents the End-Of-Sentence token. The conditional probabilities $P(w_t \mid w_1, ..., w_{t-1})$ are calculated by the model across three time steps are as follows:
   At t=1
   P(cat|<s>)=0.5, P(sat|<s>)=0.3, P(mat|<s>)=0.2 (where <s> is the Start-of-Sentence token and is not part of the vocabulary)
   At t=2
   P(sat|cat)=0.6, P(mat|cat)=0.3, P(</s>|cat)=0.1, P(cat|sat)=0.2, P(mat|sat)=0.7, P(</s>|sat)=0.1
   At t=3
   P(mat|cat sat)=0.8,P(</s>|cat sat)=0.2, P(cat|sat mat)= 0.1, P(</s>|sat mat)=0.9
   Trace the Beam Search decoding algorithm with beam width k = 2 across time steps t=1, 2, 3. At each step, compute cumulative probabilities for all expanded hypotheses and state the top-k retained candidates. Identify the final output sequence produced by the beam search. Why do we ignore P(cat|mat) and P(sat|mat)? [CO1][4+1+1=6]

3. How will you learn the sentence representation of sentence using an RNN or its variant? What is the training data for this? What is the objective function of this learning and how is the back-propagation done? Explain by designing an architecture for this. [CO3] [2+2+2=6]

4. Consider the following corpus:
   low lower low lower lowest
   low lowest low lowest
   low new
   newest widest widest
   new widest

   Apply the BPE algorithm for 2 merges. Show the vocabulary and the merge rules. How will BPE handle out-of-vocabulary words? [CO2] [2+2+2=6]

5. Consider the following 4 documents.
   Document **Text** **Class**

D1	machine learning is useful	Technology
D2	machine learning is powerful	Technology
D3	learning language is useful	Language
D4	language processing is useful	Language

Construct the vocabulary and find the TF-IDF of each word. Don’t use any start and end sentence markers. How will you represent the classes through the TF-IDF of the documents? How will you learn the representations of the documents through Doc2Vec? [Hint: For Doc2Vec, just draw the architecture of the network and write the name of the technique] [CO1+CO3] [2+2+2=6]